LING 83600: Language Technology
Uzbek ASR:
Automatic Speech Recognition through Deep Speech

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Abstract

In this paper, we explore the development of an Automatic Speech Recognition (ASR) model using Deep Speech, a deep learning-based approach for transcribing spoken language into text. The model utilizes a Short-Time Fourier Transform, convolutional and recurrent layers with 26 million parameters and is trained using a Connectionist Temporal Classification (CTC) loss function. We used the Uzbek Speech Corpus (USC) to conduct our experiments, resulting in very promising results. Through limited training, we were able to achieve a loss value of 8.4367.

1 Introduction

1.1 The Uzbek language

The Uzbek language belongs to the Turkic family of languages and is the official language of Uzbekistan. Being the most widely spoken indigenous language in Central Asia, Uzbek is estimated to have 35 to 40 million native speakers. The Uzbek language has external influences from the Arabic, Persian, and Russian languages. The writing system of the language has changed several times: from Arabic script to Cyrillic script and then to the current Latin script. Despite the official Latin script being adopted, Cyrillic script remains heavily in use. This phenomenon can be attributed to the effect of the Soviet Union's rule over Uzbekistan until 1991. The Uzbek Latin alphabet includes the standard 26 letters with the addition of "o', g', sh, ch" [10].

1.2 Inclusive Education

The drive to increase accessibility has increased more than ever in Uzbekistan. Until recently, children with disabilities were often schooled in segregated institutions [9]. The government has set a goal, to provide inclusive education in at least 51% of schools in Uzbekistan by 2025. This project is showing great success with the support of UNICEF partnerships and USAID-supported programs [3]. The schools are equipping classrooms with inclusive facilities and training educators to create a more accessible environment. This follows Uzbekistan's Concept of Development of Inclusive Education (2020-2025). Implementing inclusive education has shown to be very successful in improving the overall quality of education in countries such as Portugal and Norway.

1.3 Automatic Speech Recognition

Through the increase of technological advancements, ASR research has garnered lots of attention [11]. We have seen various ASR architectures including traditional HMM models, End-to-End Deep learning models, hybrid models that include Deep Neural Networks with Hidden Markov models, and Self-Supervised models such as Wav2Vec2 [1,4]. Numerous datasets have also been developed to conduct ASR experiments. Unfortunately, these datasets have mainly focused on popular languages such as English, Spanish, or Mandarin. This has led to underdevelopment and abandonment of low-resource languages such as Uzbek. Research in low-resourced languages is crucial not only for preserving the language but also for expanding its use in technological fields. Developing a more optimal ASR model would greatly benefit people with disabilities. Individuals with hearing loss or hard of hearing can benefit from ASR's written text and persons with mobility or vision issues can use ASR-powered voice commands to easier navigate through life.

2 Data

2.1 Challenges

One of main challenges of research in low-resource languages is the scarcity of data. Research of Natural Language Processing in the Uzbek language has been extremely limited due to the unavailability of corpora. Studies that have been done prior to 2021 have all personally crowdsourced data to be used, which were all not expansive enough to conduct meaningful research. This problem was partially addressed with the publication of the Uzbek Speech Corpus (USC) [7].

2.2 Uzbek Speech Corpus

The USC dataset is an open-source Uzbek speech corpus containing around 105 hours of transcribed audio recordings spoken by 958 different speakers from various regions and ages. To the best of our knowledge, USC is the first open-source Uzbek speech corpus available for both academic and commercial use under the Creative Commons Attribution 4.0 International License[2]. The USC was primarily designed for the task of Automatic Speech Recognition, but it can aid in other speech related tasks as well. The data gathering process consisted of crowdsourcing and audiobooks. The structure of the dataset included:

- Training Set
- Validation Set
- Testing Set

Each set contained audio recording in the format of WAV files and the corresponding transcripts as TXT files using UTF-8 encoding. The corresponding files shared the same name for easy mapping, with different extensions. Figure 1 showcases statistics present in the dataset.

2.1 Related Work

The task of creating a successful ASR model for the Uzbek language has been tackled a few times before. Musaev at al. (2020) [8] presented an End-to-End speech recognition system using Convolutional Neural Networks (CNNs), Recurrent Neural Networks (RNNs), and a Connectionist Temporal Classification (CTC) loss function. It is fascinating to read the development of this model, though the results have said to be hindered due to the size of the data used. The model is stated to have Character Error Rate (CER) of 49.1%, due to the lack of diverse speakers involved and the limited data. In the following year, another ASR model for the Uzbek language utilizing an End-to-End model was proposed [2]. The authors of this project implemented an integrated neural network model using components like CNNs ands RNNs. The CER achieved with this model was 17% which is better than the model that came before it. The dataset used for this project was very limited and the use of a larger and more diverse corpus on these models would be very interesting to see. The authors of the USC have also done experiments on their dataset and that will be the baseline for my project.

3 Baseline

3.1 Framework

For the baseline assessment of this project, I implemented one of the methods that the authors of USC conducted to test the abilities of the corpus. The study explored End-to-End (E2E) Automatic Speech Recognition (ASR) systems that were built using the ESPnet framework [6] and the WSJ recipe. Three architectures were implemented: 1) LSTM-based, 2) Transformer-based, and 3) Conformer-based. The three architectures were jointly trained under multi-task learning

Category	Train	Valid	Test	Total
Duration (hours)	96.4	4.0	4.5	104.9
Number of Utterances	100,767	3,783	3,783	108,387
Number of Words	569,000	22,500	27,100	618,600
Number of Unique Words	59,500	8,400	10,500	63,100
Speakers	879	41	38	958

Figure 1

[2] https://creativecommons.org/licenses/by/4.0/

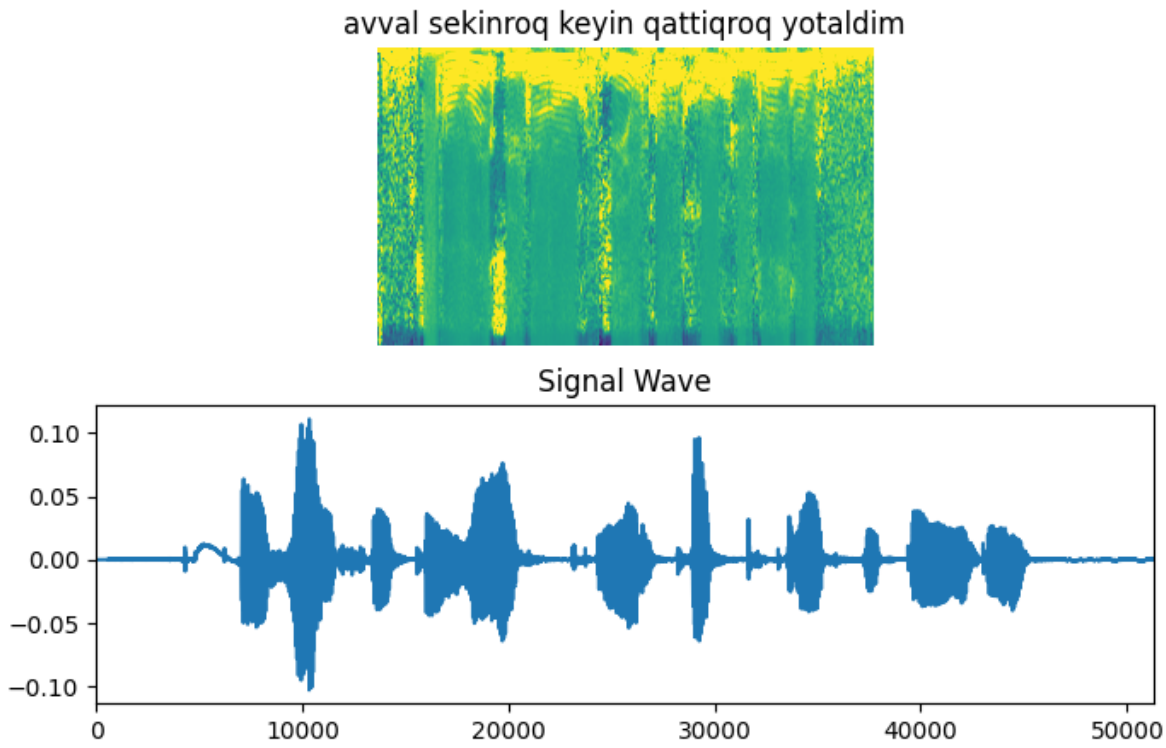


Figure 2

framework with the Connectionist Temporal Classification (CTC) objective.

3.2 Process

The input for these architectures included: speech features represented as 80-dimensional filterbank features with pitch, computer ever 10ms over. 25ms window. Outputs consisted of one 29 tokens, which represent the Uzbek alphabet. Both the Transformer and Conformer structures had encoder and decoder blocks, used self-attention and trained for a set number of epochs using the Noam optimizer.

3.3 Results

The evaluation metric that I used to measure the success of my baseline model was Word Error Rate (WER). Lower WER indicating that the ASR system is making less errors and is predicting more accurately. I will later in the paper, discuss the newly learned detriments of using WER as an evaluation metric in the Evaluation section. Using the evaluation metric of WER, the baseline model that uses E2E ASR system architecture, performed at 21.35% WER. To achieve this result, I ran recog_wav.sh with 10 different random .wav files. This shell script is made to take in a .wav file as an input and output a .txt file that should have mapped

the corresponding transcript. I compared the outputs with the labeled .txt files provided in the

dataset. After calculating the WER for each of the runs, I have averaged the results to achieve 21.35% WER.

4 Problem Setup

4.1 Data Preparation

Let a single utterance $x^{(i)}$ and label $y^{(i)}$ be samples from the training set:

$$X = \{(x^{(1)},y^{(1)}), (x^{(2)},y^{(2)}), \ldots\}$$

Each utterance $x^{(i)}$ is a time series of length $T^{(i)}$. To feed an utterance x into our model, we perform a short time Fourier transform to translate from time domain to frequency domains. This allows us to use Recurrent Neural Networks the time series nature of the data.

At each time step t, our model will provide a prediction over the set of characters, $p(l_t | x)$, where l_t =

{a, b, …, z, o', g', <*space*>, <*blank*>, <*pad*>}.

Space is used to denote word boundaries and pad is used to make all the transcriptions of equal length. The aim of this model will be to take an input sequence *x* and map it to a transcription *y*.

4.2 Model Architecture

The model we opted to use is like that of DeepSpeech2. Our model consists of several recurrent and convolution layers, and has 26 million trainable parameters.

Layer (type)	Output Shape
input (InputLayer)	(None, None, 193)
expand_dim (Reshape)	(None, None, 193, 1)
conv_1 (Conv2D)	(None, None, 97, 32)
conv_1_bn (BatchNormalization)	(None, None, 97, 32)
conv_1_relu (ReLU)	(None, None, 97, 32)
conv_2 (Conv2D)	(None, None, 49, 32)
conv_2_bn (BatchNormalization)	(None, None, 49, 32)
conv_2_relu (ReLU)	(None, None, 49, 32)
reshape (Reshape)	(None, None, 1568)
bidirectional_1 (Bidirectional)	(None, None, 1024)
dropout (Dropout)	(None, None, 1024)
bidirectional_2 (Bidirectional)	(None, None, 1024)
dropout_1 (Dropout)	(None, None, 1024)
bidirectional_3 (Bidirectional)	(None, None, 1024)
dropout_2 (Dropout)	(None, None, 1024)
bidirectional_4 (Bidirectional)	(None, None, 1024)
dropout_3 (Dropout)	(None, None, 1024)
bidirectional_5 (Bidirectional)	(None, None, 1024)
dense_1 (Dense)	(None, None, 1024)
dense_1_relu (ReLU)	(None, None, 1024)
dropout_4 (Dropout)	(None, None, 1024)
dense (Dense)	(None, None, 31)

Figure 3

We use time- and-frequency domain (2D invariance) convolution layers over time-only domain (1D invariance) since the data we are working with has components from both domains. In these layers, we apply a longer stride and wider context to speed up the training as fewer time steps are needed to model a given utterance. By downsampling the input (STFT and longer strides) we significantly speed up the model's computation, at the cost of performance.

4.3 Loss Function

Our model utilizes the Connectionist Temporal Classification (CTC) loss function.

Let *x* be an utterance, *y* be the output label, and θ being the trainable parameters of the network, we compute the loss function $L(x;y;\theta)$ and it's derivative with respect to the parameters of the network $\nabla_\theta L(x;y;\theta)$. The derivative is then used to update the network parameters through the backpropagation through time algorithm.

5 Discussion

5.1 Evaluation

In the very beginning of the training process, the model would not make any predictions unless it had a low enough loss value. On the very first epoch, it started at a loss value of about 300 and then quickly decreased as it started to gain more understanding. After that, the model started to generate bigrams. As the loss kept decreasing, we did notice that the model started to experiment with higher dimensional n-grams. As a result of this, we started seeing the model generate words that structurally look very Uzbek. The model started to recognize patterns such as a lot of adjectives in Uzbek ending in *li* or *di* so it would split words when it saw these patterns. Furthermore, in some of our later training instances, we found that the model was generating words that are Uzbek however it was not correct in the context of this sentence.

Target: *ha topdingmi dedi u anvar yaqinlashgach*
Predicted: *hartonding mi dedu onlor yaqini ish kekchib*

The example above was a very interesting instance of our training. Though none of the words predicted were correct, the model does a few very smart things. This target sentence is a difficult one as it translates roughly to: "Yes did you find him, asked Anvar as he came closer." The model recognizes that Uzbek verbs sometimes seem to end in *ing* so it splits when it sees that pattern. Even more interesting, the two words *yaqini* and *ish* are both actual Uzbek words that the model seems to recognize, though they were not used in this example. The model does struggle with spacing as there a lot of predicted transcripts will have the correct script, without the space.

In pure quantitative measurements, we were able to increase the accuracy of the model by 22%. The loss that was experienced in its lowest form was at 8.4367. The validation loss stabilized at around 75 with fluctuations present. The qualitative aspects of this model did show promising results that should only get better with more training.

5.2 Limitations

A big hinderance during this project was time. Though we had such a big dataset to work with, because we were so limited on time, we did not get to see the full potential of this model. We had a previous model that was Transformer based that we ran into a lot of different issued with which we ended up changing into this current model. Constructing the model, a lot of it from scratch, pre-processing the data took so much of the time that we were not able to train the mode to the extent that we would have liked. Additionally, ASR models are usually run on multiple GPUs but for most of our work, we were only able to use CPU and limited by 8GBs of VRAM. Later, we were able to upgrade to a GPU that used 24GBs of VRAM, we were left with limited time. We believe with a lot more training, a larger dataset and more time we would be able to have an even greater model that could potentially be ahead of the state-of-the-art model.

5.3 Lessons Learned

Through the experiments conducted in this project, we realized that though WER is a standard way of evaluation, it is certainly not the most optimal. WER is an absolute measurement as it is measuring a statistic. We believe that a relative assessment would be a better way of evaluating ASR models. A lot of aspects that our model is exceling at, cannot be shown through the sole metric of WER. As a purely quantitative assessment, WER cannot fully capture the capabilities of our model. The model is able to understand unique features of the Uzbek language that only fluent speakers of the language would be aware. It shows great success in being able to identify the traits behind Uzbek, even if it is unable to directly replicate it. Figure 4 shows an output from one of the later epochs and as you can see, the only error the model seems to make is mixing up the letter *x* and *h* which are pronounced with only slight difference.

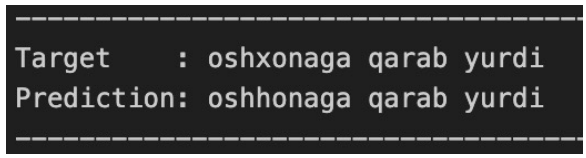


Figure 4

5.4 Future Work

In the future, we would like to spend more time to conduct further training. Due to the limited time available, we trained our model on a subset of the data available to us. We will continue to train our model after this project using the full extent of the USC and possibly more. We have also only experiment with 2D convolutions, convolutions that depend on time and frequence. We did not explore the option of using a 1D convolutions which strictly focus on time. This could be very interesting as Uzbek is not a language that puts a lot of emphasis on accents or tones.

We would also like to develop a WER that is more tuned for the Uzbek language rather than general language. Training our model with a more suitable metric would be very beneficial to its success.

5.5 Ethics

An ethical problem that is present in our project is that our model is biased towards “standard” Uzbek. Though the authors of USC claimed that the data contained native Uzbek speakers from different regions, we have found that the data is more specifically tuned to “standard” Uzbek. This could be due to the content of the speech. Since our model only focuses on one form of speaking, it underrepresents the many, different kinds of dialects that exist in Uzbekistan. There are numerous dialects that are present within the Uzbek language, but 3 main ones can be identified to include features of: Qarluq, Kipchak and Oghuz [5]. Because the data we are using is either narrations from news portals, Uzbek literature, or audiobooks, it is a very specific way of speech that we are training our model on. Though “standard” Uzbek is used in media, schools, professional settings, majority of people in Uzbekistan use their regional dialects to communicate with the people around them.

5.6 Challenges

The biggest challenge we had while working on this project was Out of Memory (OOM) issues. In the initial model that used a Transformer, we

struggled with running our code for quite some time due to OOM issues. After deciding to give up and rewriting the code from scratch to operate with a CTC loss instead of a Transformer, we were able to conduct our experiments. Due to these challenges, we did lose a lot of time and effort, but we are glad to present a model that works and possesses great potential.

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